

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Please provide project title
<i>Company:</i>	Damian Eke
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☒
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Click or tap here to enter text.
Project Background Information	
Digital exclusion is increasingly recognised as a multidimensional problem that extends beyond the availability of an internet connection. It can involve affordability, access to suitable devices, quality and reliability of connectivity, digital skills and confidence, disability and accessibility, and access to technical or social support. Consequently, conventional classifications of areas as simply 'connected' or 'unconnected' can obscure important differences in people's ability to participate meaningfully in an increasingly digital society. A number of existing initiatives, including the Digital Exclusion Risk Index (DERI) and the London Office of Technology and Innovation's Mapping Digital Exclusion Toolkit, demonstrate the value of combining publicly available datasets to identify areas at greater risk of digital exclusion. However, existing approaches generally provide a particular index or methodology rather than a flexible software toolkit through which users can explore how different indicators, datasets and weighting assumptions affect the identification of digital exclusion risk. This Year 2 Computer Science group project involves building a reusable, configurable and open-source software toolkit for analysing digital exclusion risk using publicly available UK datasets, with Nottingham as the principal case study. The toolkit will integrate datasets such as Ofcom broadband data, the ONS Postcode Directory, the English Indices of Deprivation and other appropriate open datasets relating to digital skills, demographics and socioeconomic conditions. Rather than producing a single fixed measure of digital exclusion, the toolkit will allow different indicators and weighting schemes to be explored. It will investigate how changes	

¹ An overview of different research areas in the school of computer science can be found [here](#).

in the underlying assumptions affect the geographical identification of areas at higher risk of digital exclusion. Spatial statistical techniques can then be used to identify significant geographical clusters and local patterns.

The project therefore combines software engineering, data engineering, geospatial computing, visualisation and applied statistics within a socially relevant computing problem. Nottingham will provide the case study through which the toolkit can be developed, tested and demonstrated, while the underlying software will be designed to be reusable for other UK geographical areas.

Because the project uses publicly available datasets and does not require community fieldwork or collection of personal data, it can be undertaken entirely from Nottingham. The precise indicators, research questions and technical scope will be agreed with the supervising academic team and scaled according to group size and student experience.

Requirements

- A reusable Python toolkit for analysing digital exclusion risk using configurable indicators and weighting schemes, designed to work with different UK geographical areas.
- A reproducible data pipeline that obtains, cleans, transforms and integrates publicly available datasets, principally Ofcom broadband data, the ONS Postcode Directory, the English Indices of Deprivation and selected demographic/digital-skills datasets.
- A configurable indicator system allowing different dimensions of digital exclusion risk to be included, removed or weighted differently.
- A transparent composite risk indicator, with clear documentation of the normalisation, weighting and aggregation methods used.
- Sensitivity analysis examining how different indicator selections, normalisation approaches and weighting schemes affect the resulting digital exclusion risk scores and geographical classifications.
- Spatial analysis of Nottingham, using established techniques such as Global Moran's I, Local Moran's I/LISA and, where appropriate, hotspot analysis.
- An interactive web application or dashboard allowing users to explore digital exclusion risk across Nottingham at an appropriate geographical level.
- The ability for users to change selected parameters or weighting assumptions and observe how the resulting risk map changes.
- Interactive geographical visualisations showing digital exclusion risk, individual indicators and spatial clusters.
- Automated tests covering important data-processing, indicator-calculation and spatial-analysis functions.
- Clear technical documentation, including installation instructions, data requirements, methodology and examples of how the toolkit can be applied to another UK area.
- Evidence-based analysis of Nottingham, identifying geographical patterns and examining the robustness of these patterns under different modelling assumptions.
- A reproducible software repository, using appropriate version-control and collaborative development practices.
- A final written report and presentation documenting the design, implementation, evaluation and findings of the project.

Skills	
<i>Highly desirable:</i> • Software engineering in Python (packaging, testing, documentation) • Web/application development (e.g. Flask, FastAPI, or an equivalent framework) • Data engineering / ETL • Automated software testing • Database development	<i>Desirable:</i> • Geospatial/GIS analysis (e.g. GeoPandas, QGIS) • Data visualisation • Basic statistics, for the spatial analysis components • Interest in human-centred computing and digital inclusion
Technical Proposal	
The group will develop an open-source Digital Exclusion Risk Analysis Toolkit consisting of three connected components: 1. A reusable Python analysis library 2. A reproducible public-data pipeline 3. An interactive Nottingham web application <i>The toolkit should allow a user to:</i> 1. Load relevant public datasets. 2. Validate and clean the data. 3. Harmonise geographical identifiers. 4. Calculate individual digital-exclusion indicators. 5. Normalise indicators onto comparable scales. 6. Select or modify indicator weights. 7. Generate a composite digital exclusion risk score. 8. Conduct sensitivity analysis. 9. Conduct spatial analysis. <i>Data layer:</i> The project should primarily use publicly available UK datasets, potentially including: • Ofcom Connected Nations broadband data • ONS Postcode Directory • English Indices of Deprivation • Census data • publicly available indicators relating to digital skills or internet use • other appropriate open datasets identified during the project The exact datasets should be agreed with the supervising team based on availability, licensing, geographical compatibility and data quality. <i>Toolkit layer:</i> The reusable Python package should contain separate modules for: <i>Data ingestion:</i> Functions for importing and validating supported datasets. <i>Data processing</i> Functions for: cleaning handling missing values	

geographical matching
aggregation
transformation
validation.

Normalisation: The toolkit should support an explicit and documented approach to putting different indicators onto comparable scales.

Composite indicator: The toolkit should combine the individual indicators using configurable weights.

Technical Milestones

Requirements and data investigation

- Data pipeline
- Core toolkit
- Sensitivity Analysis
- Spatial analysis
- Interactive application
- Evaluation and documentation

Submission / Award information

<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.